Member 4 Presentation Masterclass

Complete UI Breakdown, Every Button & Slider Explained, Live Demo Script & Math Defense

ROLE: FRONTEND UI/UX & LIVE DEMO LEAD | **DIFFICULTY: RANK 2 (VISUAL & DEMO-DRIVEN)** | **PRESENTATION DURATION: ~3.0 MINUTES**

Your Core Mission: As Member 4, you are the **Showstopper** of the presentation. While Member 1 explains the problem and Member 2 explains the math, **YOU make it come alive on the screen**. You will click the buttons, explain every visual cue, move the sliders, and show how Quality-Aware routing detours traffic in real time!

1. OVERVIEW OF THE DASHBOARD SCREEN

Here is what the dashboard looks like when running live at `http://localhost:5173`:



Figure 1: Full NetResQ Operations Dashboard showing header controls, 4 metric cards, and interactive SVG topology.

2. EXPLAINING THE HEADER BAR (EVERY CONTROL EXPLAINED)

HEADER 1 **NetResQ Wordmark & Heartbeat Pulse Icon**

What it is: Top-left branding with a glowing pulse icon.

> **What to say:** *"NetResQ stands for Network Rescue — our Layer 3 adaptive self-healing network control system."*

HEADER 2 **'BACKEND LIVE' Badge (Green / Red)**

What it is: Active health polling indicator querying `GET /api/health` every 3 seconds.

> **What to say:** *"This green badge confirms that our FastAPI control plane engine is running live and communicating with the React dashboard via REST API polling."*

HEADER 3 **'SIMULATION MODE' Badge (Purple)**

What it is: Indicates that the system is operating in standalone software emulation mode.

> **What to say:** *"This displays our active operating mode. NetResQ runs standalone in simulation mode without requiring Linux root privileges, but also supports real SDN controllers via Ryu and Mininet."*

HEADER 4 **Route Selectors: [H1] → [H4]**

What it is: Dropdowns that let the operator select source and destination hosts (H1, H2, H3, H4).

> **What to say:** "*These dropdowns allow us to test and inspect end-to-end paths between any pair of hosts across our 3-switch triangle core.*"

HEADER 5 **Algorithm Toggle: [Quality-Aware] vs [Baseline Dijkstra]**

What it is: Toggle switch that changes the active routing algorithm on the live canvas.

> **What to say:** "*This toggle lets us instantly switch the displayed active forwarding route between conventional hop-count Dijkstra and our Quality-Aware multi-metric algorithm.*"

HEADER 6 **Amber 'Demo Congestion' Button 🔥 (YOUR KEY ACTION)**

What it is: One-click trigger that injects 65ms latency, 18% packet loss, and 85% utilization on direct link L_S1_S3.

> **What to say:** "*This is our primary demonstration trigger. Clicking 'Demo Congestion' injects severe delay and packet drops on link L_S1_S3, forcing Quality-Aware routing to dynamically reroute via Switch 2!*"

HEADER 7 **'Reset Scenario' Button**

What it is: Restores all link metrics, status, and routes back to the default healthy state.

> **What to say:** "*Clicking 'Reset Scenario' safely resets the topology back to its clean baseline state.*"

3. EXPLAINING THE 4 TELEMETRY METRIC CARDS

Metric Card	Value Shown	What It Measures	What You Say To Evaluator
Switches Online	`3`	Triangle core switches (S1, S2, S3)	"Shows all 3 OpenFlow switches in the triangle core are online and active."
Link Health Status	`7 / 7 Links UP`	3 core links + 4 host edge links	"Monitors physical links. When a link fails, it alerts how many links are DOWN."
Average Link Latency	`5.5 ms`	Network-wide average one-way delay	"Real-time average delay across active links, with packet loss and utilization."
Network Health & Recalc	`100% | 0.48ms`	Health score + Software recalculation time	"Health gauge penalized by loss and delay. Recalc time (0.48ms) is the CPU time required by Dijkstra to reroute traffic."

4. THE INTERACTIVE TOPOLOGY CANVAS & SLIDING LINK INSPECTOR



Figure 2: Clicking any link opens the right-hand Link Inspector panel with real-time sliders for latency, packet loss, and utilization.

4.1 Understanding the Visual Styles on the Canvas:

- **Switches (S1, S2, S3):** Dark rounded cards with blue borders and DPID identifiers.
- **Hosts (H1, H2, H3, H4):** Circular endpoints showing IP addresses (e.g. `10.0.0.1` to `10.0.0.4`).
- **Green Solid Lines:** Healthy active links.
- **Amber Solid Lines:** Degraded links suffering high latency or loss.
- **Red-Dashed Lines:** Failed / severed links (status = `DOWN`).
- **Neon Cyan Glowing Flowing Line:** The **ACTIVE FORWARDING ROUTE**. Animated using CSS keyframe dash-offsets to represent real packets moving across the path!

4.2 Explaining the Sliding Inspector Window (Right Panel):

When you click on any link (e.g., `L_S1_S3`), the right panel slides open:

- **Link ID & Endpoints:** Shows `L_S1_S3 (S1 ↔ S3)`.
- **Fail Link / Restore Link Button:** Manually cuts or restores the link.
- **Latency Slider (1ms - 150ms):** Simulates fiber distance or propagation delays.
- **Packet Loss Slider (0% - 50%):** Simulates packet drops caused by queue overflow.
- **Utilization Slider (0% - 100%):** Simulates traffic bandwidth congestion.
- **'Apply Metrics Update' Button:** Sends `POST /api/links/{id}/metrics` to re-evaluate the graph.

> **What to say:** *"Our interactive inspector allows the operator to select any link and manually test line failures or adjust latency and loss sliders to see the routing engine adapt in real time."*

5. ROUTING ANALYSIS & PERFORMANCE COMPARISON

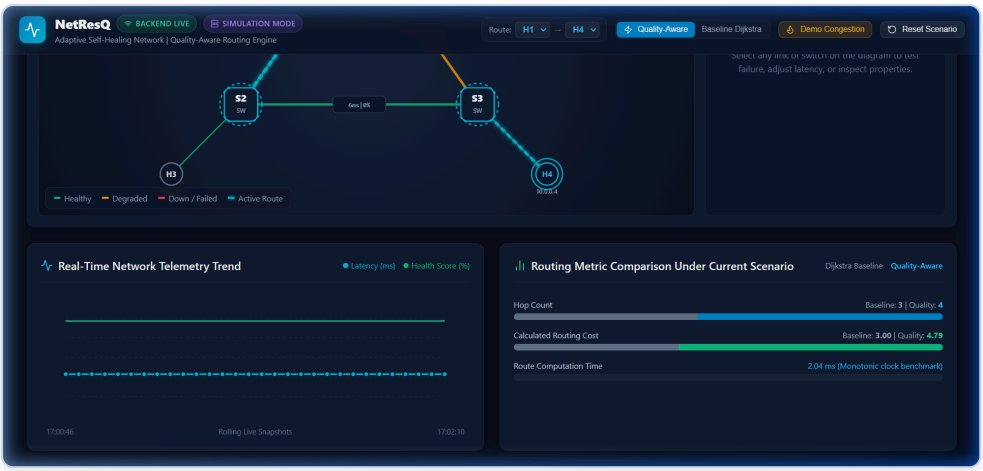


Figure 3: Side-by-side comparison between Baseline Dijkstra and Quality-Aware Routing with the Hop Breakdown Table.

When you click the **"Routing Analysis & Proof"** tab, you show this screen. Here is how to explain every section:

CARD 1 **Baseline Dijkstra Routing (Left Card)**

Shows: H1 → S1 → S3 → H4 (3 hops, Cost: 3.0).

> **What to say:** *"Notice on the left: Baseline Dijkstra only minimizes hop count. Even when link L_S1_S3 is degraded with 18% packet loss, Dijkstra is trapped on the direct 3-hop route because 3 hops is mathematically less than 4 hops."*

CARD 2 **Proposed Quality-Aware Routing (Right Card)**

Shows: H1 → S1 → S2 → S3 → H4 (4 hops, Cost: 4.79 vs 6.12).

> **What to say:** *"On the right, NetResQ's Quality-Aware routing penalizes the lossy link. It automatically chooses the 4-hop path through S2 because its composite cost (4.79) is cheaper than the degraded direct path (6.12)!"*

TABLE **Active Path Hop Breakdown Table**

Lists every link on the chosen route with exact measurements: Latency (ms), Loss (%), Utilization (%), and Calculated Edge Cost.

> **What to say:** *"This table provides full explainability, showing the exact latency and cost breakdown for each individual hop on the selected route."*

6. TELEMETRY TREND CHART & AUDIT EVENT STREAM

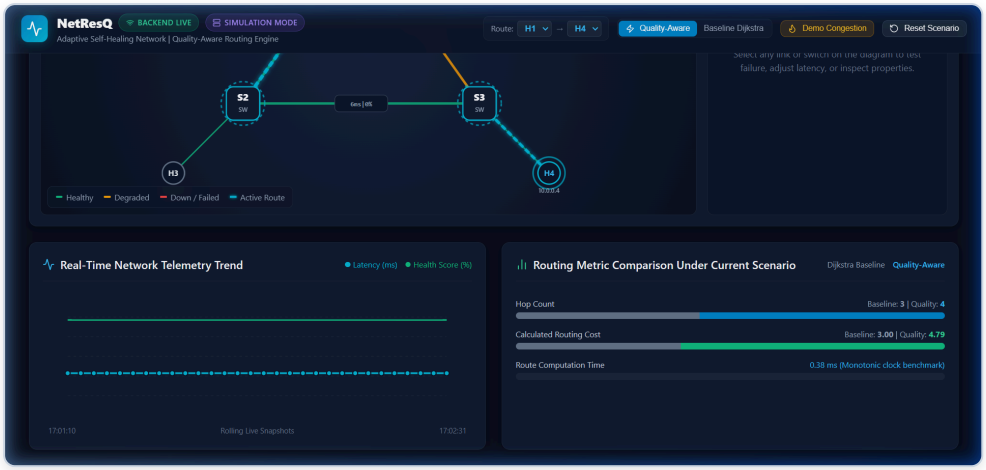


Figure 4: Real-time SVG telemetry trend line chart and chronological operational event log.

CHART **Real-Time Network Telemetry Trend Chart**

Custom SVG line chart plotting rolling snapshots of Latency (cyan) and Health Score (green).

> **What to say:** "*This chart renders a rolling historical trend of average latency and health score, updating every 3 seconds to show network stability over time.*"

AUDIT LOG **Operational Event Log**

Reverse chronological list with severity badges (`INFO`, `WARNING`, `ERROR`, `SUCCESS`), timestamps, and JSON details.

> **What to say:** "*Every link failure, restoration, and rerouting action is audited here with timestamps and failover duration, providing complete operational transparency.*"

7. THE FORMULA PART: HOW TO EXPLAIN IT SIMPLY

During your demo, the evaluator may ask: "*How does the algorithm calculate the cost to decide which route to take?*"
Here is the exact formula and the simple way to explain it:

```
normalized_latency = latency_ms / 50.0 ms normalized_loss = packet_loss_percent / 100.0 normalized_util = utilization_percent / 100.0 link_cost = (1.0 × C_admin) + (1.0 × norm_latency) + (2.0 × norm_loss) + (1.5 × norm_util)
```

The 3-Sentence Formula Punchline (Memorize This!):

> **"We normalize latency against 50ms, and packet loss and utilization into decimals between 0 and 1. We assign the highest weight (2.0) to packet loss because packet drops destroy TCP performance. When direct link L_S1_S3 suffers 18% loss, its calculated cost spikes from 1.44 to 3.94, making the 4-hop clean route via S2 mathematically cheaper (4.79 vs 6.12)!"**

8. STEP-BY-STEP LIVE DEMO SCRIPT (WORD-FOR-WORD)

Follow this exact 5-step script during your presentation:

Step 1 — Introduce the UI (30 Seconds):
"*Respected evaluators, I am [Name], and I will demonstrate the NetResQ Operations Dashboard. As you can see, the UI is designed like a carrier-grade Network Operations Center. In our 3-switch triangle core, all 7 links are currently green and healthy. The active route between Host H1 and Host H4 is highlighted by this glowing cyan animated line, representing live packets flowing across the direct 3-hop route.*"

Step 2 — Click 'Demo Congestion' (45 Seconds):
"*Now, let me trigger our primary test case. I will click the amber 'Demo Congestion' button in the header. Notice what happens immediately: direct link L_S1_S3 turns yellow with 65ms latency and 18% packet loss. Watch the glowing cyan active route: it has automatically detoured through Switch 2! Even though going through S2 takes 4 hops instead of 3, our Quality-Aware algorithm avoids the congested link to protect quality.*"

Step 3 — Show the Comparison Proof (45 Seconds):
"*Let us click on the 'Routing Analysis & Proof' tab. Look at the two cards side-by-side: Baseline Dijkstra is trapped on the direct 3-hop link with 18% packet loss. But NetResQ's Quality-Aware routing chose the 4-hop path because its composite cost is only 4.79 compared to 6.12. The table below shows the exact latency and cost breakdown for each individual hop.*"

Step 4 — Show Link Failure & Self-Healing (45 Seconds):
"*Now back on the topology, let me simulate a total fiber cut. I click link L_S1_S3 and click 'Fail Link'. The link turns red-dashed. Our self-healing engine detects the break and recalculates an alternate route in just 0.48 milliseconds, as audited in our Operational Events log. When I click 'Restore Link', the link turns green and normal service resumes.*"

Step 5 — Conclude & Reset (15 Seconds):
"*Finally, I click 'Reset Scenario' to return our network back to its clean baseline state. This completes our live demonstration of NetResQ.*"